DATA PROTECTION · EXERCISES 2B

Your first data flow diagram and your first skill

Assignment sheet for the English-language programme · Thomas More · academic year 2026–2027

In the reading module you learned what a data flow diagram is, how to read one and what a data protection officer looks for in it. In this assignment you draw one yourself, from a description of a real-looking system, and you turn what you learned into a skill: a reusable instruction file for an AI assistant. That skill is yours for the rest of the course. After every lesson you will add the criteria of that lesson to it, and at the end it will be a tool that produces, from any system description, a diagram a DPO can work with.

1. The case

CampusFit: the fitness and wellbeing app of the college

Read the description below the way a DPO would receive it from the ICT department: it is written by a technical person who knows the architecture, in plain language, without any legal terms. Everything you need for this assignment is in the text. Some facts will only become useful in later lessons; ignore them for now, but do not delete them from your diagram once you have drawn them.

What CampusFit is

The college offers all its students a free app, CampusFit, to encourage exercise and to keep an eye on their wellbeing. The app is developed and hosted by an external company, FitWorks BV, in its own data centre in the Netherlands. The college decided to introduce the app, decides which features are switched on and pays FitWorks a yearly fee.

Registration

A student logs in with the college account (single sign-on). At the first login the college's identity system sends the student's name, student number, college email address, programme and date of birth to CampusFit, which creates a profile. The student can add a profile photo and a nickname for the leaderboards.

The sports centre

At the entrance of the sports centre the student scans the student card at a turnstile. The turnstile system, managed by the facilities department, records the card number and the time of entry and sends both to CampusFit once every hour, so that the app can count sports-centre visits.

Workouts and the smartwatch

In the app a student logs workouts: type, duration and an estimate of the calories burned. Students who own a smartwatch can connect it; the watch then sends heart rate, step count and, for outdoor runs, the GPS route to CampusFit through the watch manufacturer's cloud service.

The weekly wellbeing check

Every Monday the app asks five short questions about sleep, stress and mood, with an optional free-text field. The answers are stored with the student's profile. If the score stays low for three weeks, the app suggests contacting the college's student counselling service. If the student ticks 'share with my counsellor', a counsellor receives a notification with the student's name, the scores and the free text.

Coaches and leaderboards

Sports coaches of the college log in to a coach dashboard. A coach sees the workouts and sports-centre visits of the students in his or her training group, and a leaderboard with nicknames and weekly totals that all students in the group can see.

Notifications

Reminders and coach messages are delivered as push notifications through the notification service of the phone's operating system. The notification contains the student's nickname and the text of the message.

Usage analytics

FitWorks uses an analytics tool from a US provider to understand how the app is used. For every screen a student opens, the tool receives the screen name, a timestamp, the device model, the app version and the IP address of the phone.

Research export

Once a year, the college's research group on student wellbeing receives an export. ICT replaces the student number by a random code before the export is made and keeps the list that links codes to student numbers on its own server. The export contains age, programme, gender, average workouts per week and the weekly wellbeing scores of the whole year.

Helpdesk

When a student reports a problem, the FitWorks helpdesk logs in to an administration console and can look at the full profile and history of that student. The helpdesk ticket, often with screenshots, is stored in FitWorks' ticketing system.

Logs and backups

The CampusFit server writes a log line for every request: timestamp, student number, IP address, device identifier and the action performed. Logs are kept for twelve months. FitWorks makes a full backup every night and keeps backups for ninety days.

2. What you do

Step 1 — Draw the diagram (on paper or with a tool, your choice)

- Draw the context diagram: one process for CampusFit as a whole, surrounded by every external entity you can find in the text. Label every arrow with the data it carries.
- Draw the level 0 diagram: the main processes (a verb each), the data stores (D1, D2, ...) and the flows between them. Apply the four rules from the reading module: every flow has a label, every process has input and output, no flow between two entities, no flow between two stores.
- Use whichever notation you like (Yourdon/DeMarco, Gane-Sarson, swimlanes), but use one notation consistently and make the four building blocks recognisable.

Step 2 — Mark the personal data

- Mark every flow and every store that contains personal data (a colour, a thicker line or a symbol; explain your marking in a legend).
- Mark separately, with a different symbol, the metadata: data the system creates about the use of the app rather than data the student enters. Think of timestamps, IP addresses, device identifiers, GPS routes, entry times, log lines.
- Mark the pseudonymised data and write next to it why it is still personal data.
- Write a short note (half a page) with three lists: the external parties that receive personal data, the stores that contain personal data, and anything you think is missing from the description that a DPO would want to see.

Step 3 — Write version 1 of your skill

- A skill is a text file, SKILL.md, that gives an AI assistant instructions for one recurring task. Your skill has one task: turn a plain-language system description into a data flow diagram for a DPO, or check an existing diagram against the rules.
- Use the skeleton on the next page. Fill every section in your own words, based on what you learned in the reading module: the building blocks, the levels, the four rules, the five reading steps and the personal-data criteria of this lesson (personal data, metadata, pseudonymised data).
- Keep it short and precise. An assistant follows what you write literally; vague instructions give vague diagrams.
- Give the skill an output format: which diagrams to produce, in which order, and which lists to add underneath.

Step 4 — Test your skill against your own drawing

- Give your AI assistant the skill and the CampusFit description, and ask it to produce the diagram. Do not give it your own drawing.
- Compare its result with yours. Where do they differ? Which of the two is right, and why? Did the assistant find personal data you missed, or did it miss data you found?
- Improve your skill where the comparison shows that an instruction was missing or unclear. Run the test again.
- Write a reflection of ten to fifteen lines: what did the comparison teach you about your diagram, and what about your skill?

Step 5 — Hand in

- Your context diagram and level 0 diagram (photo or export, PDF or PNG), with the legend of your markings.
- Your note from step 2.
- SKILL.md, version 1.
- The diagram your assistant produced with your skill, and your reflection from step 4.
- Name your files with your name and 'v1'. You will hand in a new version after every lesson, and the last version at the end of the course.

3. Skeleton of your skill (SKILL.md)

Copy this skeleton into a plain-text file called SKILL.md and replace every line between brackets with your own instructions. The two lines at the top (name and description) tell the assistant when to use the skill; the sections below tell it how.

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name: dfd-for-dpo
description: Turns a plain-language description of a system into a data flow diagram that a data protection officer can read, or checks an existing diagram. Use when someone asks for a DFD, a data flow diagram, or where the personal data in a system goes.
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# DFD for a DPO

## 1. Your role
(Who the assistant is, for whom it draws, and what the reader will do with the diagram.)

## 2. Building blocks and notation
(The four symbols, what each one is for, how processes, flows and stores are named and numbered. Say which notation to use and how to describe the diagram in text if the assistant cannot draw.)

## 3. Levels
(Context diagram first, then level 0; when to go to level 1.)

## 4. The four rules
(List them. Say what the assistant must do when the description breaks one of them, for example when data seems to go straight from one external party to another.)

## 5. How to read a description
(The five steps: start at the edges, follow one flow, check the rules, list the stores, ask the DPO question.)

## 6. Personal data — criteria of this lesson
(How to decide whether a flow or a store contains personal data. How to recognise metadata. Why pseudonymised data stays personal data. What to mark and how.)

## 7. Output
(What the assistant delivers, in which order: context diagram, level 0, the three lists, the questions it could not answer from the description.)

## 8. Checklist before delivering
(Short list the assistant runs through before it answers.)
```

4. How your work is assessed

Criterion	What we look at
Diagram (40%)	Correct symbols and levels; every flow labelled; the four rules respected; all external entities from the case present.
Personal data marking (25%)	Personal data, metadata and pseudonymised data correctly identified and distinguished; legend clear; the note from step 2 complete.
Skill, version 1 (25%)	Every section of the skeleton filled in, in the student's own words; instructions precise enough for an assistant to follow; output format defined.
Test and reflection (10%)	The assistant's diagram is included; the comparison is honest and specific; the skill was improved after the test.

5. Practical

- Work individually or in pairs (your lecturer will tell you which). If you work in pairs, both names go on every file.
- Tools: paper and a phone camera are fine for the diagram. If you prefer a tool: draw.io (diagrams.net) is free and has DFD shapes.
- AI assistant: any assistant that lets you paste a long instruction (Claude, ChatGPT, Copilot or another). If the assistant cannot draw, ask it to describe the diagram as lists of entities, processes, stores and flows; that is enough for the test.
- Do not paste personal data of real people into an AI assistant. The CampusFit case is fictional; keep it that way.
- Keep every version of your skill. At the end of the course you will look back at version 1 and see how far you have come.